DIINA Pilot Evaluation — Official Scoring Rubric

Educational Technology & Society · Blinded Evaluator Instrument · Version 1.0

1. Evaluation Dimensions

Dimension	Operational Definition
D1. Technical Accuracy	Factual and conceptual correctness of AI/ML content, precision of terminology, and validity of formulas and examples.
D2. Pedagogical Appropriateness	Alignment with learning objectives, instructional sequencing, clarity of explanations, and quality of assessment exercises.
D3. Linguistic Suitability	Academic register, grammatical correctness, readability, and consistency of terminology in rigorous academic English.
D4. Cultural & Contextual Relevance	Appropriateness of examples and contexts for the target learner population; sensitivity and inclusivity of content.
D5. Composite Quality	Holistic judgment of overall educational value (equal-weight mean of D1–D4).

2. Five-Point Likert Scale Anchors

Score	Anchor Description
5 – Excellent	Content is factually flawless, exemplary in pedagogy, publication-ready, culturally rich, and needs no revision.
4 – Good	Minor, non-substantive flaws only; high-quality educational content with negligible revision needed.
3 – Adequate	Generally correct and usable, but with notable gaps in depth, clarity, or contextual fit.
2 – Poor	Frequent errors or weak pedagogical structure; substantial revision required before educational use.
1 – Unacceptable	Fundamentally incorrect, misleading, inappropriate, or unusable; total rewrite required.

3. Scoring Criteria and Procedure

1. Independent rating. Each evaluator rates every item under both conditions (Baseline and DIINA) independently, without knowledge of the generation condition, using the anchors above.

2. Dimension scores. Assign one integer score (1–5) for each of Dimensions D1–D4 per item per condition. Half-points are not permitted.

3. Composite score. D5 is computed as the equal-weight arithmetic mean of D1–D4, rounded to two decimals; it is used for primary paired comparisons.

4. Rationale. For any score of 2 or lower, or 5, a brief written justification is required in the audit-log column of the evaluation matrix.

5. Aggregation. Item-level scores are averaged across the five evaluators per condition; reliability is assessed with ICC(2,1) (two-way random, absolute agreement) and reported with 95% confidence intervals.

6. Interpretation bands. Item mean < 2.5 = revise; 2.5–3.49 = acceptable with revision; 3.5–4.24 = good; ≥ 4.25 = exemplary.

7. Conflict resolution. Discrepancies of ≥ 2 points between evaluators on any dimension are flagged and discussed in a consensus session before final aggregation.

Scoring is recorded in **raw/DIINA_Pilot_Evaluation_Matrix.xlsx**; cleaned ratings and derived statistics are stored in **processed/** and **results/** respectively (see metadata/README.md).